

TABLE 2
COMPARISON OF GRID BATTERY WITH PROPOSED HVDC CORRIDOR

Parameters	Life (Years)	Investment	Dependability	Rare Earth Elements	Use	Degradation in Capacity	Losses (%)
Grid Battery	5-10*	High + Replacement cost	Dependable and fast response	Required in large quantities	Storage only	Storage capacity degrades*	5 to 15*
Proposed Power Corridor	>50	High + Maintenance cost	Daily and seasonal variations	No such requirement	Storage + Power transmission	No such degradation	7 to 15 ^

*Depends upon depth and frequency of discharge; ^ Depends upon length

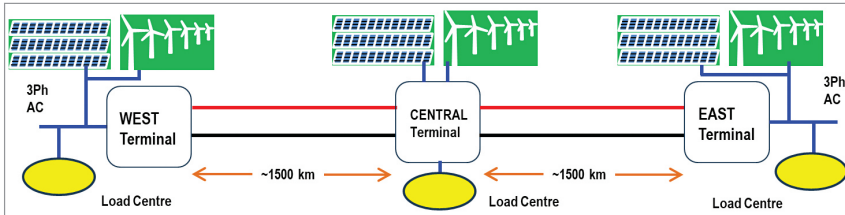


Fig. 3: Three-terminal power corridor

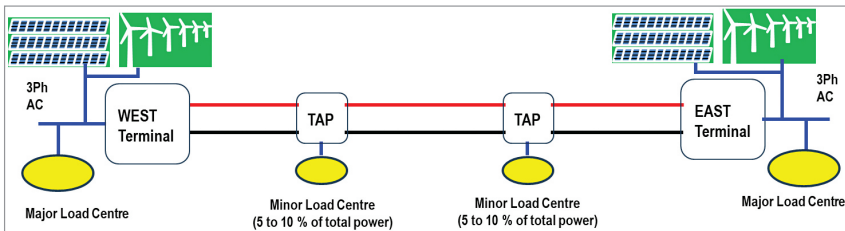


Fig. 4: Two-terminal power corridor with taps for intermediate small load centres

shows the performance comparison of grid battery with the HVDC corridor.

Implementation

Four examples of the proposed power corridor are given here. There may be many more possibilities based on the detailed study of the region where it must be implemented.

1. Two-terminal corridor. Fig. 2 is an example of a two-terminal corridor. Here, the total distance between eastern and western terminals is covered using a single HVDC link. It will provide maximum time difference between the two geographical locations.

2. Three-terminal corridor. Fig. 3 shows a three-terminal power corridor. In this scheme, one extra terminal is introduced in the middle of the corridor. The 3000km transmission line is now divided into HVDC lines of 1500km each. The central terminal

should have potential to generate solar and wind energy. Also, a load centre should be present nearby to consume power.

In such a scheme, time difference will be about one hour from the central terminal on either side. If required, power can be transmitted directly from west to east terminals with a small increase in power losses. Such a corridor will provide more flexibility, and power availability will be more uniform throughout 24 hours.

3. Corridors with tappings. Fig. 4 shows a two-terminal corridor with a few taps introduced along the HVDC line. With advancements in HVDC technology, it is now possible to tap small amounts of power from the DC link. Up to 20% of power can be tapped. Beyond 20%, it becomes difficult to control the HVDC link. Such taps are useful for providing power to small load centres in rural areas

located directly on the power corridor.

4. Augmentation of existing HVDC lines. We have several HVDC lines in use in our country. A detailed study of renewable power generation

on these existing lines should be conducted. This study will provide clues as to which existing HVDC lines could be extended in the east-west direction. These are good candidates for expanding renewable power generation while utilising existing infrastructure.

Relevance

As the share of renewables in the grid increases, there is a need for adding battery storage at the grid level. Table 2 clarifies that the proposed power corridor can also act as a battery. It also offers additional advantages. Therefore, the proposed corridor also becomes relevant in the present scenario. When the link is idle, it can be used for transmitting power from conventional power plants. During emergencies, we can utilise this infrastructure.

As the share of renewables in the grid increases, there is a need to install grid batteries. These batteries help in stabilising the grid. An east-west power corridor with an HVDC link can extend PV power generation due to the time difference between east and west terminals. It acts as a battery and provides several advantages. It is necessary to evaluate this alternative for additional benefits. Hence, while planning newer HVDC links, the proposed power corridor should be considered as another possible solution. Furthermore, we can explore existing HVDC links for extension in the east-west direction to take advantage of the time difference in solar energy generation. **EFY**